1 Ghost Trace Test

This document tests whether incremental dirty rect rendering leaves ghost traces after editing. Open the built-in preview, then perform the edits below.

1.1 Test 1: Delete a middle list item

1. First item with enough text to wrap across multiple lines so the paragraph is wide and fills a substantial portion of the page width for testing dirty rect coverage when content is deleted and surrounding text reflows upward.

2. **DELETE THIS ENTIRE LINE** – second item that also contains substantial text wrapping across multiple lines at a different horizontal position than the items above and below it in the list for triggering the dirty rect edge-case bug.

3. Third item with more content filling several lines to ensure we have enough vertical space for the dirty rect algorithm to encounter rows with varying horizontal pixel positions after the middle item is removed from the document.

1.2 Test 2: Delete a blockquote

Normal paragraph at full page width that is positioned above the blockquote. When the quote below is deleted, this text stays in place but the text below it will shift upward, creating a dirty region that spans different horizontal positions.

> This blockquote is indented from both sides. When this entire block is deleted, the paragraph below will move up into this position. The dirty rect must cover the full width of the quote text AND the moved paragraph text, which start at different horizontal offsets – triggering the edge-case bug if the rect width is computed incorrectly in the single-pass loop.

Another normal paragraph at full page width. When the blockquote above is deleted, this text shifts upward. The area where the blockquote was (indented) is now filled with this full-width text. The dirty rect algorithm must handle the varying min_x values across rows correctly.

1.3 Test 3: Rapid undo/redo

Type some random characters here, then undo them. Watch for any visual artifacts or ghost traces that remain after the undo operation completes.

Another normal paragraph at full page width. When the blockquote above is deleted, this text shifts upward. The area where the blockquote was (indented) is now filled with this full-width text. The dirty rect algorithm must handle the varying min_x values across rows correctly.

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$$a = b + c \tag{1a}$$
$$d = e - f \tag{1b}$$
$$g = h \times i \tag{1c}$$
$$j = k \div l \tag{1d}$$
$$p = \sqrt{q} \tag{1e}$$
$$r = \log(s) \tag{1f}$$
$$t = \sin(u) \tag{1g}$$
$$v = \cos(w) \tag{1h}$$
$$\tag{1i}$$